

Dr. Puneet Kumar

M.Tech. | Ph.D. | Computational Biologist & Drug Discovery Scientist

Project Scientist, Department of Biosciences and Bioengineering, IIT Roorkee, India

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Publications: 14

Citations: 136

h-index: 8

Total IF = 37.7

PROFESSIONAL SUMMARY

Computational Biologist with a Ph.D. from IIT Roorkee and over seven years of research experience at the interface of molecular simulations, structure-based drug design, and AI-driven drug discovery. My work focuses on decoding protein conformational dynamics and ligand-binding mechanisms to identify and optimise small-molecule therapeutics — with a track record of 14 peer-reviewed publications (h-index: 8, 136 citations) across computational chemistry, cancer biology, inflammation, and neurodegeneration. I bring hands-on proficiency in the full in silico drug discovery pipeline: from virtual screening and molecular docking (Schrödinger Maestro) to free energy calculations (FEP, QM/MM) and biomolecular MD simulations (GROMACS, Desmond). I am also a co-inventor of a patented IoT biosensor (IIT Roorkee IPR Cell) and a 2026 Sakura Science Exchange Program Awardee (Hokkaido University, Japan).

RESEARCH INTERESTS

- Machine learning and AI for drug discovery and biomedical data analysis
- Computational and systems biology of cancer, inflammation, and neurodegeneration
- Molecular dynamics simulations and structure-based drug design
- Translational bioinformatics, precision medicine, and target identification
- Computational target identification and structure-activity relationship (SAR) analysis

EDUCATION

Ph.D. in Biosciences and Bioengineering	2019 – 2024
Indian Institute of Technology Roorkee (IIT Roorkee), India — Funded by Ministry of Education, Government of India	
Thesis: Computational Approaches Toward Understanding Molecular Mechanism of Ligands Targeting Integrin Receptors	CGPA = 8.54
Supervisor: Professor Pranita P. Sarangi	
Integrated M.Tech. in Biomedical Engineering	2013 – 2018
Indian Institute of Information Technology Allahabad (IIIT Allahabad), India	
Thesis: Gene Expression Analysis of Oral Sub-mucous Fibrosis in Association with Dysplasia and Oral Squamous Cell Carcinoma	CGPA = 7.93
Supervisor: Professor Sintu Kumar Samanta	

PROFESSIONAL EXPERIENCE

Project Scientist

Oct 2024 – Present

Department of Biosciences and Bioengineering, IIT Roorkee, India	
Leading interdisciplinary computational drug discovery projects integrating molecular dynamics, machine learning, and transcriptomic data analysis to identify novel therapeutic targets in inflammation and cancer.	
Applying DFT, QM/MM, and ML approaches to characterise transcriptomic signatures of systemic inflammation and identify small-molecule candidates.	
Co-inventor of a tail-mounted IoT biosensor for precision livestock monitoring (patent filed through IIT Roorkee IPR Cell); developing AI-based decision-support systems for climate-controlled dairy farming.	
Mentoring junior researchers and coordinating multi-disciplinary research activities across biology, engineering, and clinical science teams.	
Bioinformatics Expert	Jun 2018 – Nov 2018
Center of Excellence in Pharmaceutical Sciences (CEPS), GGS Indraprastha University, New Delhi, India	
Served as the computational lead within a multidisciplinary team of medicinal chemists, mycologists, and pharmaceutical scientists for an antifungal drug discovery programme targeting novel nucleoside-based compounds.	
Conducted structure-based virtual screening and molecular docking using Schrödinger Maestro against fungal Lanosterol 14α-demethylase (CYP51) — a clinically validated antifungal drug target.	
Integrated computational insights with experimental medicinal chemistry to support rational lead identification and optimisation.	
Research Intern	May 2016 – Jul 2016
Terrablue XT, Bengaluru, India	
Contributed to early-stage research and development activities in a health-tech startup environment.	

RESEARCH & TECHNICAL SKILLS

Molecular Simulations: GROMACS, Desmond, metadynamics, mixed-solvent MD, induced-fit MD, biomolecular dynamics; conformational free energy landscape analysis

Structure-Based Drug Design (SBDD): Molecular docking, induced-fit docking, virtual screening, free energy perturbation (FEP), QM/MM calculations — primarily using the Schrödinger Maestro suite

Ligand-Based Drug Design: QSAR modelling, pharmacophore modelling, structure-activity relationship (SAR) analysis

AI & Computational Methods: Machine learning integrated with SBDD workflows; ML applied to transcriptomic and multi-omics data for target identification and biomarker discovery; DFT calculations

Programming & Data Analysis: Python, R, Grace, Microsoft Excel; Arduino for IoT sensor programming and data integration

Other: Hardware-software integration for decision-support systems; large-scale continuous data stream handling for AI-based health monitoring

PUBLICATIONS

denotes equal contribution

1. **Kumar, P.**, Sharma, A., & Sarangi, P. P. (2026). Transcriptomic signature of systemic inflammation by integration of machine learning, DFT, and QM/MM approach. (Under revision)

2. **Kumar, P.#**, Umar, M.#, Naeem, A., & Khan, E. (2026). AI in designing novel cancer treatments. In *Artificial Intelligence for Enhanced Diagnosis in Oncology* (pp. 149-168). Academic Press. [\[Link\]](#)
3. **Kumar, P.#**, Gupta, S.#, & Sarangi, P. P. (2025). Fibulin-7 and its bioactive fragments: emerging immunomodulatory roles in disease. *Inflammation Research*, 74, 101. (IF = 5.4) [\[Link\]](#)
4. **Kumar, P.#**, Sharma, A.#, Islam, A., Bhardwaj, M., Kumar, V., & Prakash, H. (2025). The neuroprotective role of chlorogenic acid and Fisetin in differentiated neuronal cell line-SHSY5Y against amyloid- β -induced neurotoxicity. *Toxicology in Vitro*, 106110. (IF = 2.7) [\[Link\]](#)
5. **Kumar, P.**, Kumar Rai, S., & Sarangi, P. P. (2024). An in-silico study reveals that a C-terminal fragment of the adhesion protein Fibulin7 (Fbln7-C) regulates the activation of integrin $\alpha 5\beta 1$ through dynamics of VWA and the hybrid domain in the $\beta 1$ subunit. *Journal of Biomolecular Structure and Dynamics*, 1–20. (IF = 2.7) [\[Link\]](#)
6. **Kumar, P.**, Sharma, P., Singh, D., Mishra, N., & Sarangi, P. P. (2024). Unraveling the molecular basis for effective regulation of integrin $\alpha 5\beta 1$ for enhanced therapeutic interventions. *Biochemical and Biophysical Research Communications*, 734, 150627. (IF = 2.2) [\[Link\]](#)
7. Dalpati, N., Rai, S. K., Dash, S. P., **Kumar, P.**, Singh, D., & Sarangi, P. P. (2024). Integrins $\alpha 5\beta 1$ and $\alpha v\beta 3$ Differentially Participate in the Recruitment and Reprogramming of Tumor-associated Macrophages in the In Vitro and In Vivo Models of Breast Tumor. *The Journal of Immunology*. (IF = 3.4) [\[Link\]](#)
8. **Kumar, P.#**, Dipankar, P.#, & Sarangi, P. P. (2023). In silico identification and characterization of small-molecule inhibitors specific to RhoG/Rac1 signaling pathway. *Journal of Biomolecular Structure and Dynamics*, 41(2), 560–580. (IF = 2.7) [\[Link\]](#)
9. **Kumar, P.#**, Dipankar, P.#, Dash, S. P., & Sarangi, P. P. (2021). Functional and therapeutic relevance of Rho GTPases in innate immune cell migration and function during inflammation: An in-silico perspective. *Mediators of Inflammation*, 2021(1), 6655412. (IF = 4.2) [\[Link\]](#)
10. Chakraborty, P., Dalpati, N., Bhan, C., Dash, S. P., **Kumar, P.**, & Sarangi, P. P. (2020). A C-terminal fragment of adhesion protein Fibulin7 regulates neutrophil migration and functions and improves survival in LPS induced systemic inflammation. *Cytokine*, 131, 155113. (IF = 3.7) [\[Link\]](#)
11. Chakraborty, P., Dash, S. P., Dalpati, N., **Kumar, P.**, Jain, D., & Sarangi, P. P. (2021). A C-terminal fragment of adhesion protein fibulin-7 inhibits growth of murine breast tumor by regulating macrophage reprogramming. *The FEBS Journal*, 288(3), 803–817. (IF = 4.2) [\[Link\]](#)
12. Bhan, C., Dash, S. P., Dipankar, P., **Kumar, P.**, Chakraborty, P., & Sarangi, P. P. (2019). Investigation of extracellular matrix protein expression dynamics using murine models of systemic inflammation. *Inflammation*, 42, 2020–2031. (IF = 5.0) [\[Link\]](#)
13. Mishra, D., Panda, G., **Kumar, P.**, & Singh, S. (2017). Novel drug delivery system for herbal formulation in cancer treatment. *World Journal of Pharmaceutical Research*, 6, 341–353. (IF = 0.5) [\[Link\]](#)
14. **Kumar, P.**, Mishra, S., & Singh, S. (2017). Advanced acuity in microbial biofilm genesis, development, associated clinical infections and control. *Journal des Anti-infectieux*, 19(1), 20–31. (IF = 0.5) [\[Link\]](#)
15. Mishra, S., **Kumar, P.**, & Singh, S. (2017). Structural analysis of protein lysyl oxidase: modelling and simulation study. *Journal of Biotech Research*, 8, 9. (IF = 0.5) [\[Link\]](#)

AWARDS & HONORS

- 2026: Sakura Science Exchange Program Awardee — Hokkaido University, Japan
- 2019: Institute Assistantship, IIT Roorkee
- 2018: GATE Qualified (Graduate Aptitude Test in Engineering)
- 2018: Teaching Assistantship, IIIT Allahabad
- 2017: Silver Medal — Researcher's Day Poster Presentation, IIIT Allahabad

CONFERENCES & PRESENTATIONS

- 2025: Oral Presentation for the ISAGB Young Researcher Award — ISAGBCON-2025, Kolkata, India
- 2023: Oral Presentation — IMMUNOCON 2023, New Delhi, India
- 2022: Poster Presentation — Biotech Day, IIT Roorkee, India

INVITED EXPERT LECTURES

- May 2024: Invited Speaker, ICAR-CIRB, Hisar, India. National Workshop on Artificial Intelligence and Sensors for Efficient Livestock Farming.
- Dec 2025: Invited Speaker, AICB, New Delhi, India. Leveraging Artificial Intelligence for Organizational Development.

EDITORIAL ACTIVITIES & ACADEMIC SERVICES

Editorial Appointments

- Associate Editorial Board Member, *Current Biological Sciences* (June 2026 - Present)
- Topic Co-ordinator, *Frontiers in Oncology* (June 2026 - Present)

Peer Review Contributions

- Discover Oncology
- Heliyon (Cell Press)
- International Journal of Biological Macromolecules
- Biochemical and Biophysical Research Communications
- Journal of Biosciences

REFERENCES

- Dr. Raksha Sharma (Associate Professor) — Dept. of Computer Science and Engineering, IIT Roorkee | raksha.sharma@cs.iitr.ac.in
- Dr. Nidhi Mishra (Associate Professor) — Dept. of Applied Sciences, IIIT Allahabad | nidhimishra@iiita.ac.in
- Dr. Hridayesh Prakash (Professor & Deputy Director) — Amity Centre for Translational Research, Amity University Noida | hprakash@amity.edu

Dr. Puneet Kumar

Date: 3rd June 2026 | Place: Roorkee, Uttarakhand